

WhatsApp CRM

Product Requirements Document

A WhatsApp-first CRM for shared inbox, lead capture, deals, tasks, notes, and sales visibility

Field	Value
Version	v1.0
Date	May 7, 2026
Prepared For	MaVoid Internship Program
Product Type	WhatsApp-first sales CRM
Team Structure	UI/UX, Frontend, Backend / Full Stack, DevOps
Delivery Window	21 days
Document Status	Final PRD for MVP execution

Document Contents

1. 1. Executive Summary
2. 2. Product Strategy and Positioning
3. 3. Users, Roles, and Permissions
4. 4. MVP Scope and Non-Scope
5. 5. WhatsApp Platform Rules and Constraints
6. 6. Core User Flows
7. 7. Functional Requirements
8. 8. Screen Requirements
9. 9. Data Model
10. 10. API Requirements
11. 11. UX and Design Requirements
12. 12. Technical Requirements
13. 13. DevOps and Deployment Requirements
14. 14. Timeline and Internship Responsibilities
15. 15. Acceptance Criteria
16. 16. Final Demo Script
17. 17. Risks and Mitigations
18. 18. Reporting and Evaluation
19. 19. Future Backlog
20. 20. References

1. Executive Summary

MaVoid WhatsApp CRM is a WhatsApp-first sales CRM that centralizes customer conversations, leads, contacts, deals, tasks, notes, and dashboard visibility into one internal operating system.

The product is not a normal CRM with WhatsApp added later. WhatsApp is the main input channel, the inbox is the main workspace, and CRM records are created from or linked to WhatsApp conversations.

The MVP should replace manual business communication scattered across WhatsApp chats with a shared, assigned, searchable, trackable system.

1.1 Document Snapshot

Field	Value
Product Name	MaVoid WhatsApp CRM
Primary User	Small sales team using WhatsApp for customer communication
Main Workflow	Incoming WhatsApp message -> conversation -> lead/contact/deal -> task/note -> dashboard
MVP Goal	Build a working WhatsApp shared inbox connected to basic CRM modules
Hard Constraint	No advanced CRM features until WhatsApp inbox works
Success Definition	A user can receive a WhatsApp message, reply from CRM, create a lead/deal, add follow-up task, and view dashboard metrics

1.2 Why This Product Exists

- Most customer conversations already happen on WhatsApp, but operational visibility is weak.
- Manual WhatsApp chats hide ownership, follow-ups, deal progress, and customer context.
- Sales teams need a shared inbox and CRM record system, not another disconnected spreadsheet.
- The product becomes a reusable MaVoid demo for operational software, automation, and internal systems.

1.3 Product Principle

Build a small product that works. Do not build a fake enterprise monster. This project fails if the team chases Salesforce-level features before the inbox, webhook, message sending, and CRM linking are stable.

2. Product Strategy and Positioning

2.1 Strategic Fit

The product supports MaVoid's public positioning as a company that builds AI automation, software, and operational systems for enterprises. It demonstrates process control, lead routing, reporting, visibility, and centralized execution.

2.2 Positioning Statement

MaVoid WhatsApp CRM centralizes sales conversations, leads, deals, tasks, and follow-ups into one operational system instead of letting business activity disappear inside WhatsApp chats.

2.3 Product Goals

- Provide one shared WhatsApp inbox for business conversations.
- Create CRM records from WhatsApp contacts and conversations.
- Assign conversations and deals to sales users.
- Track lead status and deal pipeline progress.
- Create follow-up tasks so sales opportunities do not disappear.
- Show simple dashboard metrics for sales visibility.
- Deploy a real working app with Docker, CI/CD, and webhook support.

2.4 Non-Goals

- Do not build a full WhatsApp Web clone.
- Do not replace personal WhatsApp usage.
- Do not support WhatsApp groups, calls, status updates, or personal contacts.
- Do not build campaign automation in MVP.
- Do not build a chatbot in MVP.
- Do not build complex multi-tenant SaaS billing.
- Do not build advanced sales forecasting or custom report builder.

3. Users, Roles, and Permissions

3.1 User Personas

Persona	Main Need	Primary Screens	Success Outcome
Admin	Configure system, manage users, monitor all data	Settings, Users, Inbox, Dashboard	System runs and all sales activity is visible
Sales Manager	Supervise conversations, assign owners, track pipeline	Inbox, Leads, Deals, Tasks, Dashboard	No lead is ignored and team progress is visible
Sales Rep	Reply to customers, qualify leads, manage own deals	Inbox, CRM side panel, Leads, Deals, Tasks	Customers receive replies and follow-ups happen on time

3.2 Role Permission Matrix

Capability	Admin	Sales Manager	Sales Rep
Login/logout	Yes	Yes	Yes
View WhatsApp settings	Yes	No	No
Edit WhatsApp settings	Yes	No	No
View all conversations	Yes	Yes	No
View assigned conversations	Yes	Yes	Yes
Reply to assigned conversations	Yes	Yes	Yes
Reply to unassigned conversations	Yes	Yes	No
Assign conversations	Yes	Yes	No
Close/reopen conversations	Yes	Yes	Assigned only

Capability	Admin	Sales Manager	Sales Rep
Create users	Yes	No	No
View all leads	Yes	Yes	No
View own leads	Yes	Yes	Yes
Create leads	Yes	Yes	Yes
Delete leads	Yes	No	No
View all deals	Yes	Yes	No
View own deals	Yes	Yes	Yes
Move deal stage	Yes	Yes	Own only
View dashboard	Global	Global/team	Own data

4. MVP Scope and Non-Scope

4.1 MVP Modules

Module	Priority	Description
Authentication	P0	Login, logout, current user, protected routes, role guards
User Management	P0	Admin creates users and assigns roles
WhatsApp Integration	P0	Store account config, verify webhook, send/receive messages
Shared Inbox	P0	Conversation list, chat thread, assignment, statuses, unread counts
Conversations	P0	Open/pending/closed conversations tied to WhatsApp contacts
Messages	P0	Inbound/outbound text messages, statuses, webhook logs
Templates	P0	Manual local approved-template records and template sending
WhatsApp Contacts	P0	Auto-create WhatsApp contact from inbound message
Leads	P0	Create and qualify leads from conversations
Contacts	P0	Store customer person records linked to WhatsApp contact
Companies	P0	Store company records linked to contacts/deals
Deals	P0	Basic sales pipeline with stages and value
Tasks	P0	Follow-up tasks linked to leads/deals/conversations
Notes	P0	Internal notes linked to CRM records
Dashboard	P0	Conversation, lead, deal, and task metrics
Docker + CI/CD	P0	Local and deployed working system

4.2 MVP Must Not Include

Feature	Reason
WhatsApp groups	Unsupported or unsuitable for CRM MVP workflow
WhatsApp calls	Not required for sales CRM workflow
WhatsApp status	Not part of shared inbox CRM

Feature	Reason
AI chatbot	High risk and unnecessary for first delivery
Broadcast campaigns	Compliance, cost, and operational complexity
Bulk import	Useful later, not needed for first working product
Multi-number inbox	Adds account routing complexity
Advanced analytics	Dashboard metrics are enough
Payment collection	Different product scope
Mobile app	Responsive web only

4.3 Scope Control Rule

Any feature that does not directly support receiving a WhatsApp message, replying, linking CRM data, tracking follow-up, or viewing basic performance is deferred.

5. WhatsApp Platform Rules and Constraints

The product must respect WhatsApp Business Platform behavior. These rules shape product design, backend logic, and user expectations.

Constraint	Product Impact	MVP Requirement
Webhooks are required	Incoming messages and status updates must enter through webhook events	Implement GET/POST webhook endpoints and WebhookLog records
Service window limits free-form replies	Users cannot always type arbitrary outbound messages	Composer must show whether free reply is allowed or template is required
Templates are required for business-initiated messages outside allowed window	Template sending flow is mandatory for practical usage	Manual approved-template records and variable fill UI
Opt-in is required for template messaging	System must avoid careless outbound messaging	Contact should include opt-in flag and opt-in source field
Pricing applies to delivered template messages	Dashboard should track template sends in future	MVP stores template message status; cost estimate is P1
Webhook retries can duplicate events	Same message may arrive more than once	Deduplicate by WhatsApp message ID
Production number risk	Interns should not touch real MaVoid number first	Use Meta test number before production

5.1 Mandatory Platform Assumptions

- Use WhatsApp Business Platform / Cloud API or an approved Business Solution Provider only.
- Backend must expose a public HTTPS webhook URL before real inbound testing.
- The system must store raw webhook payloads for debugging.
- Message sending must be abstracted behind a backend WhatsApp service layer.
- Access tokens must be treated as secrets and never exposed to frontend.
- Test number first. Production number later.

6. Core User Flows

6.1 Main WhatsApp-to-CRM Flow

21. Customer sends WhatsApp message to business number.
22. Meta sends webhook event to backend.
23. Backend logs raw payload in WebhookLog.
24. Backend identifies or creates WhatsAppContact by phone number.
25. Backend creates or updates Conversation.
26. Backend stores inbound Message.
27. Conversation appears in shared inbox.
28. Manager/admin assigns conversation to Sales Rep.
29. Sales Rep replies from CRM.
30. Sales Rep creates or links lead/contact/deal from side panel.
31. Sales Rep adds note and follow-up task.
32. Dashboard metrics update.

6.2 Reply Flow

33. User opens assigned conversation.
34. System checks if free-form reply is allowed.
35. If allowed, user sends text reply.
36. If not allowed, user selects approved template.
37. Backend sends message via WhatsApp API.
38. System stores outbound message with status sent/pending.
39. Webhook status updates message to delivered/read/failed.

6.3 Lead Creation Flow

40. User opens conversation side panel.
41. User clicks Create Lead.
42. System pre-fills name/phone/source from WhatsAppContact.
43. User assigns owner and status.
44. Lead is linked to WhatsAppContact and Conversation.
45. Lead appears in Leads module and dashboard metrics.

7. Functional Requirements

7.1 Authentication and Users

ID	Requirement	Acceptance Criteria
AUTH-001	User can log in with email and password.	Valid credentials return JWT/session. Invalid credentials show clear error. Inactive users cannot log in.
AUTH-002	User can log out.	Session/token is cleared. User returns to login page. Protected routes become inaccessible.
AUTH-003	Frontend can fetch current user.	GET /auth/me returns id, name, email, role, status. Invalid token returns 401.

ID	Requirement	Acceptance Criteria
AUTH-004	Role-based access enforced on frontend and backend.	Unauthorized API access fails even if frontend button is hidden.
USER-001	Admin can create users.	Name, email, password, role, status are saved. Password is hashed. Email is unique.
USER-002	Admin can update users.	Admin can update name, role, and status. Audit-friendly updatedAt changes.

7.2 WhatsApp Settings

ID	Requirement	Acceptance Criteria
WA-SET-001	Admin can view WhatsApp connection settings.	Settings page shows business name, phone number, WABA ID, phone number ID, webhook status, integration status.
WA-SET-002	Admin can save WhatsApp API configuration.	Access token is stored encrypted or through environment secret. Token is never returned in full after save.
WA-SET-003	Admin can send a test message.	System sends a test template or text message when rules allow. Result is logged.
WA-SET-004	Webhook verification endpoint works.	GET /webhooks/whatsapp validates verify_token and returns challenge when valid.

7.3 Webhook Processing

ID	Requirement	Acceptance Criteria
WEB-001	Backend receives WhatsApp webhook events.	POST /webhooks/whatsapp accepts message and status events.
WEB-002	Backend logs every webhook payload.	WebhookLog stores raw payload, event type, processed flag, error message when applicable.
WEB-003	Inbound messages create/update conversations.	New phone number creates WhatsAppContact and Conversation. Existing number updates same conversation.
WEB-004	Inbound messages are deduplicated.	Same WhatsApp message ID is not stored twice.
WEB-005	Status webhooks update outbound messages.	Message status becomes sent, delivered, read, or failed based on status event.
WEB-006	Unsupported message types are handled safely.	System stores unsupported placeholder and does not crash.

7.4 Shared Inbox and Conversations

ID	Requirement	Acceptance Criteria
INBOX-001	Users can view conversation list.	List shows customer name/phone, last message, unread count, assigned user, status, last message time.
INBOX-002	Users can filter conversations.	Filters include all, mine, unassigned, unread, open, pending, closed.
INBOX-003	Users can open conversation thread.	Thread shows inbound/outbound messages, timestamp, direction, delivery/read/failure status.
INBOX-004	Manager/admin can assign conversation.	Assigned user changes and appears immediately in conversation list and side panel.
INBOX-005	Users can change conversation status.	Open, pending, and closed statuses are supported.
INBOX-006	Unread count is tracked.	Inbound messages increment unread count. Opening or marking read clears count for authorized user.

7.5 Message Sending

ID	Requirement	Acceptance Criteria
MSG-001	User can send text reply when allowed.	Text is sent through backend WhatsApp service. Outbound message appears in thread.
MSG-002	Composer shows reply availability.	UI displays Free reply allowed, Template required, Sending, or Failed state.
MSG-003	Failed messages show retry state.	Failed outbound messages display failure status and retry option if feasible.
MSG-004	Backend stores all outbound messages.	Message record includes conversation, direction, type, body, sentByUserId, status, timestamp.
MSG-005	Media messages are safely represented.	MVP may show media placeholder. Text messages are P0; media preview is P1.

7.6 Templates

ID	Requirement	Acceptance Criteria
TPL-001	Admin/manager can create local template records.	Template stores name, category, language, body, variables, status.
TPL-002	User can select template from composer.	Template modal displays body preview and required variable fields.
TPL-003	User can send template message.	Backend sends template through WhatsApp API and stores outbound template message.
TPL-004	Template variables are validated.	All required variables must be filled before sending.
TPL-005	Template approval is outside CRM.	CRM only stores templates already approved in Meta/BSP dashboard.

7.7 WhatsApp Contacts and CRM Linking

ID	Requirement	Acceptance Criteria
WAC-001	System auto-creates WhatsAppContact.	New inbound phone creates record with phone and profile name if available.
WAC-002	User can link WhatsAppContact to Lead.	Linked lead appears in side panel and lead detail links back to conversation.
WAC-003	User can link WhatsAppContact to Contact.	Linked CRM contact appears in side panel.
WAC-004	User can create Lead from conversation.	Lead form is prefilled with name, phone, source=WhatsApp.
WAC-005	User can create Deal from conversation.	Deal can link to lead/contact/company and assigned owner.

7.8 Leads

ID	Requirement	Acceptance Criteria
LEAD-001	User can create lead.	Lead requires name or phone, owner, status. Default status is New.
LEAD-002	Users view leads based on permission.	Admin/Manager see all. Rep sees own assigned leads.
LEAD-003	Lead list supports search, filter, pagination.	Search by name, phone, email/company. Filter by status, owner, source.
LEAD-004	User can update lead status.	Statuses: New, Contacted, Qualified, Unqualified.
LEAD-005	Lead detail shows related conversation, notes, tasks, deals.	All linked context is visible on detail page.

7.9 Contacts and Companies

ID	Requirement	Acceptance Criteria
CONTACT-001	User can create CRM contact.	Name and owner are required. Phone/email optional but recommended.
CONTACT-002	Contact can link to company.	Contact detail shows company and related deals/notes.
COMP-001	User can create company.	Name and owner are required. Industry, size, website, location optional.
COMP-002	Company detail shows linked contacts and deals.	Related lists are visible and searchable when enough data exists.

7.10 Deals

ID	Requirement	Acceptance Criteria
DEAL-001	User can create deal.	Title, value, owner, and stage are required. Default stage is Prospecting.
DEAL-002	Deals can be viewed as pipeline.	Columns: Prospecting, Qualified, Proposal, Negotiation, Won, Lost.
DEAL-003	Deal stage can be updated.	Stage update persists and dashboard metrics update.
DEAL-004	Deal detail shows CRM context.	Shows company, contact, owner, value, stage, expected close date, notes, tasks.
DEAL-005	Pipeline value is calculated.	Open pipeline excludes Won and Lost unless specifically requested.

7.11 Tasks

ID	Requirement	Acceptance Criteria
TASK-001	User can create follow-up task.	Task requires title, due date, assigned user. Can link to conversation, lead, or deal.
TASK-002	Tasks support pending/done status.	User can mark task done.
TASK-003	Overdue state is computed.	Pending task with due date before today appears overdue.
TASK-004	Task list supports filters.	Filter by status, assigned user, due date, related record.

7.12 Notes

ID	Requirement	Acceptance Criteria
NOTE-001	User can add note.	Note content is required. Note shows author and timestamp.
NOTE-002	Notes can link to multiple CRM parent types.	MVP supports exactly one parent per note: conversation, lead, contact, company, or deal.
NOTE-003	Users can view notes on detail pages.	Notes appear in chronological order.
NOTE-004	Delete note follows permissions.	Admin can delete any. Manager team notes. Rep own notes.

7.13 Dashboard

ID	Requirement	Acceptance Criteria
DASH-001	Dashboard shows conversation metrics.	Open, unread, assigned, unassigned, messages sent today.
DASH-002	Dashboard shows lead metrics.	Total leads, new leads, qualified leads, leads by status.
DASH-003	Dashboard shows deal metrics.	Open deals, won deals, pipeline value, deals by stage.

ID	Requirement	Acceptance Criteria
DASH-004	Dashboard shows task metrics.	Overdue tasks and upcoming tasks.
DASH-005	Dashboard respects permissions.	Admin global. Manager global/team. Rep own assigned data.

8. Screen Requirements

8.1 Login

- Logo/brand name
- Email field
- Password field
- Login button
- Loading state
- Invalid credentials error
- Responsive layout

8.2 WhatsApp Inbox

This is the most important screen. If this screen is weak, product is weak.

Area	Required Elements
Left conversation list	Customer name/phone, preview, unread count, assigned user, status, timestamp, filters
Center chat thread	Message bubbles, direction, timestamp, status, unsupported placeholder, loading state
Message composer	Free-text input, template selector, reply availability state, send button, failure handling
Right CRM panel	WhatsApp contact, linked lead/contact/deal, owner, notes, tasks, actions

8.3 CRM Side Panel Actions

- Assign conversation
- Create lead
- Link existing lead
- Create contact
- Create company
- Create deal
- Add note
- Create task
- Close conversation

8.4 WhatsApp Settings

- Business name
- Phone number
- WABA ID
- Phone number ID
- Webhook verification status

- Integration status
- Test message button
- Secret/token handling notice

8.5 Templates

- Templates list
- Create/edit local template record
- Category
- Language
- Body preview
- Variable fields
- Send test button

8.6 Leads

- Leads table
- Create lead button
- Search
- Status filter
- Owner filter
- Source filter
- Lead detail
- Related conversation link

8.7 Deals

- Pipeline kanban
- Deal cards
- Create deal
- Stage filters
- Deal detail
- Notes/tasks
- Won/lost states

8.8 Dashboard

- Open conversations
- Unread conversations
- Messages sent today
- New WhatsApp leads
- Qualified leads
- Open deals
- Pipeline value
- Overdue tasks
- Deals by stage chart
- Leads by status chart

9. Data Model

9.1 Entity Overview

Entity	Purpose
User	System user with role and status
WhatsAppAccount	Connected WhatsApp business configuration
WhatsAppContact	External WhatsApp sender/customer identity
Conversation	Inbox thread tied to WhatsApp contact
Message	Inbound/outbound WhatsApp message record
MessageStatusEvent	Delivery/read/failure event history
WhatsAppTemplate	Local copy of approved WhatsApp template
WebhookLog	Raw webhook debugging log
Lead	Potential customer or sales inquiry
Contact	CRM person record
Company	CRM organization record
Deal	Sales opportunity
Task	Follow-up work item
Note	Internal context record

9.2 Key Entity Fields

Entity	Important Fields
User	id, name, email, passwordHash, role, status, createdAt, updatedAt
WhatsAppAccount	id, businessName, phoneNumber, phoneNumberId, wabalid, accessTokenEncrypted/secretRef, verifyToken, webhookSecret, status
WhatsAppContact	id, phone, profileName, optInStatus, optInSource, linkedLeadId, linkedContactId
Conversation	id, whatsappContactId, assignedUserId, status, unreadCount, lastMessageAt, lastMessagePreview, customerServiceWindowExpiresAt
Message	id, conversationId, whatsappMessageId, direction, type, body, mediaUrl, status, sentByUserId, timestamp
MessageStatusEvent	id, messageId, status, rawPayload, createdAt
WhatsAppTemplate	id, name, category, language, body, variablesJson, status
WebhookLog	id, eventType, rawPayload, processed, errorMessage, createdAt
Lead	id, name, email, phone, source, status, assignedOwnerId, whatsappContactId, conversationId
Contact	id, name, email, phone, jobTitle, companyId, ownerId, whatsappContactId
Company	id, name, industry, website, size, location, ownerId
Deal	id, title, companyId, contactId, value, stage, ownerId, expectedCloseDate
Task	id, title, description, dueDate, status, assignedToId, conversationId, leadId, dealId
Note	id, content, authorId, conversationId, leadId, contactId, companyId, dealId

9.3 Enumerations

Enum	Values
Role	ADMIN, SALES_MANAGER, SALES_REP
ConversationStatus	OPEN, PENDING, CLOSED
MessageDirection	INBOUND, OUTBOUND
MessageType	TEXT, IMAGE, DOCUMENT, AUDIO, VIDEO, TEMPLATE, UNSUPPORTED
MessageStatus	RECEIVED, SENT, DELIVERED, READ, FAILED
TemplateCategory	UTILITY, MARKETING, AUTHENTICATION
TemplateStatus	ACTIVE, INACTIVE
OptInStatus	UNKNOWN, OPTED_IN, OPTED_OUT
LeadStatus	NEW, CONTACTED, QUALIFIED, UNQUALIFIED
DealStage	PROSPECTING, QUALIFIED, PROPOSAL, NEGOTIATION, WON, LOST
TaskStatus	PENDING, DONE

9.4 Relationship Rules

- One WhatsAppContact can have one active Conversation for MVP.
- Conversation belongs to one WhatsAppContact.
- Conversation can link to Lead, Contact, Deal through side panel actions.
- Lead can link to WhatsAppContact and Conversation.
- Contact can link to WhatsAppContact and Company.
- Deal can link to Company, Contact, owner, notes, and tasks.
- Task can link to Conversation, Lead, or Deal.
- Note can link to exactly one parent record for MVP to avoid messy polymorphic ambiguity.

10. API Requirements

10.1 API Standards

- All protected endpoints require authenticated user.
- All list endpoints support pagination.
- Errors return consistent response shape.
- Swagger/OpenAPI documentation is required.
- Backend must never expose WhatsApp access tokens to frontend.

10.2 Success Response Format

```
{ "success": true, "data": {} }
```

10.3 List Response Format

```
{ "success": true, "data": [], "pagination": { "page": 1, "limit": 10, "total": 50, "totalPages": 5 } }
```

10.4 Error Response Format

```
{ "success": false, "message": "Invalid credentials", "error": "AUTH_INVALID_CREDENTIALS" }
```

10.5 Endpoint Inventory

Module	Endpoints
Auth	POST /auth/login; GET /auth/me; POST /auth/logout
Users	GET /users; POST /users; GET /users/:id; PATCH /users/:id
WhatsApp Webhook	GET /webhooks/whatsapp; POST /webhooks/whatsapp
WhatsApp Settings	GET /settings/whatsapp; PATCH /settings/whatsapp; POST /settings/whatsapp/test-message
Conversations	GET /conversations; GET /conversations/:id; PATCH /conversations/:id/assign; PATCH /conversations/:id/status; PATCH /conversations/:id/read
Messages	GET /conversations/:id/messages; POST /conversations/:id/messages/text; POST /conversations/:id/messages/template
Templates	GET /whatsapp/templates; POST /whatsapp/templates; GET /whatsapp/templates/:id; PATCH /whatsapp/templates/:id; DELETE /whatsapp/templates/:id
WhatsApp Contacts	GET /whatsapp-contacts; GET /whatsapp-contacts/:id; PATCH /whatsapp-contacts/:id/link
Leads	GET /leads; POST /leads; GET /leads/:id; PATCH /leads/:id; DELETE /leads/:id
Contacts	GET /contacts; POST /contacts; GET /contacts/:id; PATCH /contacts/:id; DELETE /contacts/:id
Companies	GET /companies; POST /companies; GET /companies/:id; PATCH /companies/:id; DELETE /companies/:id
Deals	GET /deals; POST /deals; GET /deals/:id; PATCH /deals/:id; PATCH /deals/:id/stage; DELETE /deals/:id
Tasks	GET /tasks; POST /tasks; GET /tasks/:id; PATCH /tasks/:id; DELETE /tasks/:id
Notes	GET /notes; POST /notes; DELETE /notes/:id
Analytics	GET /analytics/summary; GET /analytics/conversations; GET /analytics/leads-by-status; GET /analytics/deals-by-stage; GET /analytics/overdue-tasks
Webhook Logs	GET /webhook-logs; GET /webhook-logs/:id

10.6 Query Parameters

Endpoint	Supported Query Parameters
GET /conversations	page, limit, search, status, assignedUserId, unread, mine, unassigned
GET /messages	page, limit, before, after
GET /leads	page, limit, search, status, ownerId, source
GET /contacts	page, limit, search, ownerId, companyId
GET /companies	page, limit, search, ownerId, industry, size
GET /deals	page, limit, search, stage, ownerId, companyId
GET /tasks	page, limit, status, assignedToId, dueDateFrom, dueDateTo, overdue
GET /notes	conversationId, leadId, contactId, companyId, dealId

11. UX and Design Requirements

11.1 Design Direction

- Serious enterprise dashboard style.
- Blue/cyan MaVoid brand direction.
- Clean tables, readable chat layout, practical forms.
- No playful startup clutter.
- No fake futuristic AI robot visuals.
- Use whitespace and hierarchy to reduce cognitive load.

11.2 Brand Tokens

Token	Value	Usage
Primary Blue	#105FA1	Main buttons, links, active navigation
Secondary Cyan	#5DB1D4	Highlights, gradients, chart accents
Electric Accent	#00FFC6	Small status accents only
Dark Surface	#0A192F	Sidebar/dark panels
Dark Section	#0C1E36	Premium surfaces
Light Surface	#F1F5F9	Page background/cards
Font	Josefin Sans / fallback sans-serif	Headings and UI where available

11.3 Required UI States

State	Where Required	Notes
Loading	Inbox, tables, dashboards, forms	Use skeletons or clear loading indicators
Empty	Inbox, leads, deals, tasks, notes	Explain what user should do next
Error	API failure, login, sending messages	Show clear retry path
Success	Form submit, message send, assignment	Toast or visible feedback
Disabled	Unauthorized actions, invalid forms	Avoid confusing clickable controls
Failed message	Chat thread	Message bubble must show failure state
Template required	Composer	User must understand why free text is blocked

11.4 Responsive Requirements

Viewport	Requirement
Desktop	Full three-column inbox: conversation list, chat thread, CRM side panel
Tablet	Conversation list and chat remain usable; side panel can collapse
Mobile	Basic inbox and reply capability; CRM side panel becomes drawer or separate tab

12. Technical Requirements

12.1 Frontend Stack

Area	Technology
Framework	Next.js
Language	TypeScript
Styling	Tailwind CSS
Components	shadcn/ui
Forms	React Hook Form
Validation	Zod
Data Fetching	React Query or equivalent
Charts	Recharts
Realtime/Polling	Polling for MVP; WebSockets optional later

12.2 Backend Stack

Area	Technology
Framework	NestJS
Database	PostgreSQL
ORM	Prisma
Auth	JWT
Password Hashing	bcrypt or argon2
Validation	DTOs with class-validator or equivalent
Docs	Swagger/OpenAPI
WhatsApp Client	Backend service wrapping Meta Cloud API or BSP API

12.3 Backend Critical Rules

- Deduplicate incoming messages by WhatsApp message ID.
- Log raw webhook payload before processing.
- Never trust frontend for permission checks.
- Do not expose access tokens or secrets.
- Use transaction-safe creation when inbound message creates contact, conversation, and message.
- Return consistent error responses.
- Add seed data for demo users and sample CRM records.

13. DevOps and Deployment Requirements

13.1 Required DevOps Outputs

- Frontend Dockerfile

- Backend Dockerfile
- docker-compose.yml with PostgreSQL
- .env.example
- GitHub Actions CI pipeline
- Public HTTPS backend URL for webhook
- Live frontend URL
- Live API/Swagger URL
- Deployment guide
- Database backup and restore commands
- Log access notes

13.2 Environment Variables

Variable	Purpose
DATABASE_URL	PostgreSQL connection string
JWT_SECRET	JWT signing secret
JWT_EXPIRES_IN	Token expiry
FRONTEND_URL	Allowed frontend origin
BACKEND_URL	Public backend URL
WHATSAPP_ACCESS_TOKEN	Meta/BSP API token; secret only
WHATSAPP_PHONE_NUMBER_ID	Meta phone number ID
WHATSAPP_WABA_ID	WhatsApp Business Account ID
WHATSAPP_VERIFY_TOKEN	Webhook verification token
WEBHOOK_SECRET	Webhook validation secret if used
NODE_ENV	development/staging/production

13.3 CI/CD Requirements

46. Install frontend dependencies.
47. Install backend dependencies.
48. Lint frontend.
49. Lint backend.
50. Build frontend.
51. Build backend.
52. Run tests where available.
53. Build Docker images or validate Docker build.

14. Timeline and Internship Responsibilities

14.1 21-Day Delivery Plan

Days	Phase	Team Deliverable
0	Setup	Repo, Figma, project board, group, report template, deployment target

Days	Phase	Team Deliverable
1-2	Product Definition	WhatsApp CRM flow, API contract, webhook architecture, Figma structure
3-4	Design System + Skeleton	Inbox UI design, app shell, backend skeleton, Docker base, webhook verification
5-6	Auth + Users	Login, roles, users, protected routes, seed users, CI pipeline
7-8	Webhook + Inbox Foundation	Receive inbound messages, log payload, create conversation, show inbox
9-10	Send Replies	Send text replies, store outbound messages, status updates, failed state
11-12	CRM Linking	Create lead/contact/deal from WhatsApp conversation side panel
13-14	Templates	Template records, variable fill UI, template send, customer window indicator
15-16	Tasks + Dashboard	Follow-up tasks, notes, WhatsApp dashboard metrics
17-21	QA + Final Demo	Deployed app, working webhook, demo flow, README, final Loom

14.2 Track Responsibilities

Track	Owns	Final Deliverables
UI/UX	Product flows, inbox UX, CRM side panel, template modal, responsive layouts, design system	Figma file, flows, screen designs, component library, UX audit
Frontend	Next.js app, inbox UI, chat thread, CRM side panel, forms, dashboard, API integration	Working frontend, reusable components, responsive UI, final demo
Backend / Full Stack	NestJS API, database, auth/RBAC, webhooks, WhatsApp API wrapper, CRM logic	Working APIs, Swagger docs, seed data, webhook processing, Postman collection
DevOps	Docker, CI/CD, deployment, HTTPS webhook, secrets, logs, backups	Docker setup, live app/API, pipeline, deployment guide, backup script

15. Acceptance Criteria

15.1 Product Acceptance

- Admin can log in and create users.
- Webhook verification works on public backend URL.
- Inbound WhatsApp message creates contact, conversation, and message records.
- Conversation appears in shared inbox.
- Authorized user can reply from CRM.
- Outbound message appears in chat thread.
- Message status updates are stored from webhook events.
- User can create lead from WhatsApp conversation.
- User can create deal from conversation or linked contact.
- User can add note and task.
- Dashboard shows real conversation, lead, deal, and task metrics.
- Docker Compose runs locally.
- CI pipeline passes.
- Final deployed demo works without fake data dependency.

15.2 UI/UX Acceptance

- Figma is organized and named clearly.

- Inbox screen has desktop and mobile behavior.
- CRM side panel is clear and usable.
- Template modal includes variable fields.
- Dashboard is readable and not overdecorated.
- Design follows MaVoid brand direction.

15.3 Frontend Acceptance

- All major routes exist.
- Protected routes work.
- Inbox can display conversations and messages from API.
- Message composer supports text/template states.
- Forms validate and show errors.
- Loading, empty, error, and success states exist.
- Responsive layout does not break.

15.4 Backend Acceptance

- Auth and role guards work.
- Webhook GET/POST work.
- Inbound message handling is idempotent.
- Message status updates work.
- All PO APIs are documented in Swagger.
- Seed data exists.
- WebhookLog captures raw payloads and processing errors.

15.5 DevOps Acceptance

- Docker Compose starts frontend/backend/database.
- Public HTTPS backend URL is available for Meta webhook.
- CI/CD pipeline passes.
- Environment variable documentation is complete.
- Deployment guide is clear enough for another developer to follow.
- Backup and restore commands are documented and tested.

16. Final Demo Script

54. Login as Admin.
55. Open WhatsApp settings and show webhook connected.
56. Receive inbound WhatsApp message from test number.
57. Show conversation appears in inbox.
58. Assign conversation to Sales Rep.
59. Login/open view as Sales Rep.
60. Reply from CRM.
61. Show outbound message in chat thread.
62. Create lead from conversation.
63. Qualify lead.

64. Create contact/company if needed.
65. Create deal from CRM side panel.
66. Add note to deal/conversation.
67. Create follow-up task.
68. Send template message when composer requires template.
69. Open dashboard and show updated metrics.
70. Show webhook logs and message status record.

17. Risks and Mitigations

Risk	Impact	Mitigation
WhatsApp setup delays	Inbox cannot be tested	Use Meta test number early; set up webhook by Day 4
Production number misuse	Business communication disruption	Never use real number until staging passes
Webhook debugging confusion	Backend stalls	WebhookLog is mandatory from first webhook test
Duplicate webhook events	Duplicate messages	Deduplicate by WhatsApp message ID
Template rule confusion	Users think replies are broken	Composer must show Free reply allowed vs Template required
Scope creep	No finished product	Cut broadcasts, chatbot, multi-number, advanced analytics
Frontend waits for backend	Lost time	Backend publishes API contract by Day 2
DevOps late deployment	No working webhook	Public backend URL required by Day 7

18. Reporting and Evaluation

18.1 Individual 48-Hour Report

Each trainee submits this every 48 hours:

- Name:
- Track:
- Report Number:
- Date:
- Current Phase:
- Assigned Feature:
- Completed Work:
- Proof: Figma / GitHub PR / Live URL / Screenshots / Loom
- Blockers:
- Dependencies: Need from UI/UX / Frontend / Backend / DevOps
- Next 48 Hours:
- Self Rating: Execution /10, Communication /10, Quality /10

18.2 Evaluation Criteria

Track	Criteria
UI/UX	CRM product thinking 20, inbox UX 20, visual quality 20, responsive design 15, design system 15, communication 10
Frontend	UI accuracy 20, inbox implementation 20, code structure 15, API integration 20, responsive/error states 15, communication 10

Track	Criteria
Backend	Webhook/API quality 25, data model 15, auth/RBAC 15, WhatsApp integration 20, docs/testing 15, communication 10
DevOps	Docker 20, CI/CD 20, deployment/webhook availability 25, secrets/env 15, logs/backup 10, documentation 10

19. Future Backlog

Feature	Priority After MVP	Notes
Broadcast campaigns	Later	Requires opt-in, template governance, rate/cost controls
AI reply suggestions	Later	Can be useful after message history and CRM context are stable
Auto-assignment rules	Later	Assign by working hours, lead source, or rep capacity
Media previews	P1/P2	Images/documents/audio support
Multi-number inbox	Later	Requires routing by account/phoneNumberId
Advanced analytics	Later	Conversion rate, response time, deal velocity
CSV import/export	Later	Useful but not MVP-critical
Conversation reminders	Later	Notify reps about idle conversations
Full activity timeline	P1	Events across lead/deal/task/message changes

20. References

External platform rules should be checked again before production launch because Meta can change WhatsApp Business Platform behavior and pricing.

- Meta - WhatsApp Business Platform overview - [official documentation](#)
- Meta - Webhooks overview - [official documentation](#)
- Meta - Messages webhook reference - [official documentation](#)
- Meta - Service messages / send messages - [official documentation](#)
- Meta - Template fundamentals - [official documentation](#)
- Meta - Get opt-in for WhatsApp - [official documentation](#)
- Meta - WhatsApp Business Platform pricing - [official documentation](#)
- WhatsApp Business Messaging Policy - [official documentation](#)

21. Final Scope Lock

Build WhatsApp shared inbox + basic CRM. Nothing more until this works end-to-end.

- Receive WhatsApp message.
- Show conversation in shared inbox.
- Reply from CRM.
- Create lead/deal/task/note.
- Show dashboard metrics.
- Deploy with working webhook.